

TONGJI CALCULUS, 7TH EDITION · CHAPTER 1

FUNCTIONS & LIMITS EXERCISE WORKBOOK

100 original and textbook-method adaptations · basic to challenge

Goodnotes-optimized 4:3 edition · 2026



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How to use this set

100 Questions	All 10 sections Sections	Basic → challenge Progression	8 mixed formats Formats
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Work through Foundation, Methods, Synthesis, and Challenge in that order. Each question shows its difficulty, format, and suggested time. Attempt it before opening the solution book; record the cause of each error and retry after 48 hours.

“Textbook-method adaptation” means the representative method is retained while the structure, parameters, or task have been independently redesigned.

Scope: Chapter 1 tools only. Derivatives, L'Hopital's rule, Taylor expansions, and power series are not used.

Coverage matrix

1	Mappings and Functions	11	domains, composition, inverses, properties
2	Limits of Sequences	10	epsilon-N, properties, subsequences, recursion
3	Limits of Functions	10	one-sided limits, epsilon-delta, properties
4	Infinitesimals and Infinite Quantities	7	infinitesimals, infinity, reciprocals
5	Algebra of Limits	13	factoring, rationalization, parameters
6	Existence Criteria and Two Fundamental Limits	16	squeeze, monotone-bounded, two limits
7	Comparison of Infinitesimals	11	orders, equivalentents, cancellation traps

Sec.	Topic	Items	Key coverage
8	Continuity and Discontinuities	10	continuity, parameters, classification
9	Operations on Continuous Functions	4	operations, composition, elementary functions
10	Continuous Functions on Closed Intervals	8	extrema, zeros, IVT, uniform continuity

Foundation

Attempt independently before consulting the solutions.

§1 · Mappings and Functions

Basic

Single choice

4 min

Q001 · Identify a Valid Mapping

Let $X=\{-1,0,1\}$ and $Y=\{0,1\}$. Which correspondence defines a function from X to Y ?

- A. A. For every $x \in X$, define $f(x)=x^2$
- B. B. Define $f(-1)=1$ and $f(1)=1$, but do not define $f(0)$
- C. C. Define $f(-1)=1$ and $f(0)=0$, while assigning both $f(1)=0$ and $f(1)=1$
- D. D. For every $x \in X$, define $f(x)=x+1$

WORKSPACE · add pages as needed

§1 · Mappings and Functions

Basic

Fill in the blank

5 min

Textbook-method adaptation

Q002 · Common Domain of a Radical and a Logarithm

The natural domain of $f(x)=\sqrt{2-x}+\ln(x+1)$ is _____.

WORKSPACE · add pages as needed

§1 · Mappings and Functions

Basic

True/false with justification

3 min

Q003 · A Necessary Condition for Composition

True or false: If $(f \circ g)(x_0)$ is defined, then $g(x_0)$ must belong to the domain of f .

WORKSPACE · add pages as needed

§2 · Limits of Sequences

Basic

Single choice

5 min

Textbook-method adaptation

Q004 · Recognize the Rigorous Definition of a Sequence Limit

Which epsilon-N statement correctly expresses that the sequence $\{a_n\}$ converges to A ?

- A. A. For every $\varepsilon > 0$, there exists a positive integer N such that $|a_n - A| < \varepsilon$ whenever $n > N$
- B. B. There exists a positive integer N such that, for every $\varepsilon > 0$, $|a_n - A| < \varepsilon$ whenever $n > N$
- C. C. For every $\varepsilon > 0$ and every positive integer n , $|a_n - A| < \varepsilon$
- D. D. There exists $\varepsilon > 0$ such that, for every positive integer N , one can find $n > N$ with $|a_n - A| < \varepsilon$

WORKSPACE · add pages as needed

§2 · Limits of Sequences

Basic

Fill in the blank

6 min

Textbook-method adaptation

Q005 · Choose N for a Prescribed Accuracy

For $a_n = (-1)^n/n$, the limit is _____. When $\varepsilon = 0.01$ and the definition uses $n > N$, one may take $N =$ _____.

WORKSPACE · add pages as needed

§2 · Limits of Sequences	Basic	True/false with justification	4 min
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Q006 · Does Boundedness Guarantee Convergence?

True or false: Every bounded sequence converges.

WORKSPACE · add pages as needed

§3 · Limits of Functions

Basic

Single choice

5 min

Textbook-method adaptation

Q007 · Does a Point Value Affect a Function Limit?

Define $f(1)=7$, and for $x \neq 1$ let $f(x)=(x^2-1)/(x-1)$. What is $\lim_{x \rightarrow 1} f(x)$?

- A. A. 0
- B. B. 1
- C. C. 2
- D. D. 7

WORKSPACE · add pages as needed

§3 · Limits of Functions

Basic

True/false with justification

4 min

Q008 · Are Two Existing One-Sided Limits Sufficient?

True or false: If $\lim_{x \rightarrow a^-} f(x)$ and $\lim_{x \rightarrow a^+} f(x)$ both exist and are finite, then $\lim_{x \rightarrow a} f(x)$ must exist.

WORKSPACE · add pages as needed

§4 · Infinitesimals and Infinite Quantities

Basic

Single choice

4 min

Q009 · Recognizing an Infinitesimal

As $x \rightarrow 0$, which of the following functions is an infinitesimal?

- A. A. $2+x$
- B. B. $x^2/(1+|x|)$
- C. C. $1/x^2$
- D. D. $1+\sin x$

WORKSPACE · add pages as needed

§4 · Infinitesimals and Infinite Quantities

Basic

True/false with justification

5 min

Q010 · Is an Infinitesimal a Small Number?

True or false, with justification: An infinitesimal is a nonzero number whose absolute value is very small.

WORKSPACE · add pages as needed

§5 · Algebra of Limits	Basic	Single choice	4 min
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Q011 · When Can Limits Be Divided Directly?

Which conclusion follows directly from the quotient limit law?

- A. $f(x) \rightarrow 2$ and $g(x) \rightarrow 3$, so $f(x)/g(x) \rightarrow 2/3$
- B. $f(x) \rightarrow 2$ and $g(x) \rightarrow 0$, so $f(x)/g(x) \rightarrow 0$
- C. $f(x) \rightarrow 0$ and $g(x) \rightarrow 0$, so $f(x)/g(x) \rightarrow 1$
- D. Neither $f(x)$ nor $g(x)$ has a limit, so $f(x)/g(x)$ must have no limit

WORKSPACE · add pages as needed

§5 · Algebra of Limits

Basic

Calculation

4 min

Q012 · Direct Substitution for a Polynomial

Evaluate $\lim_{x \rightarrow 2} (3x^3 - 2x^2 + x - 5)$.

WORKSPACE · add pages as needed

§5 · Algebra of Limits

Basic

Calculation

6 min

Textbook-method adaptation

Q013 · Removing a Zero Factor with a Difference of Cubes

Evaluate $\lim_{x \rightarrow 2} (x^3 - 8)/(x - 2)$ without derivatives.

WORKSPACE · add pages as needed

§6 · Existence Criteria and Two
Fundamental Limits

Basic

Single choice

4 min

Q014 · Recognizing the Complete Squeeze-Theorem Hypotheses

Which set of conditions is sufficient to conclude $f(x) \rightarrow L$ by the squeeze theorem?

- A. A. $g(x) \leq f(x) \leq h(x)$, with $g(x) \rightarrow L$ and $h(x) \rightarrow L$
- B. B. $g(x) \leq f(x)$, with $g(x) \rightarrow L$
- C. C. $f(x) \leq h(x)$, with $h(x) \rightarrow L$
- D. D. $g(x) \leq f(x) \leq h(x)$, but $g(x) \rightarrow L_1$ and $h(x) \rightarrow L_2$ with $L_1 < L_2$

WORKSPACE · add pages as needed

§6 · Existence Criteria and
Two Fundamental Limits

Basic

Calculation

5 min

Textbook-method adaptation

Q015 · Suppressing Oscillation with a Vanishing Factor

Evaluate $\lim_{x \rightarrow 0} x \sin(1/x)$.

WORKSPACE · add pages as needed

§6 · Existence Criteria and
Two Fundamental Limits

Basic

Fill in the blank

4 min

Textbook-method adaptation

Q016 · Scaling the First Fundamental Limit

$\lim_{x \rightarrow 0} \sin(5x)/(3x) = \underline{\hspace{2cm}}$.

WORKSPACE · add pages as needed

§7 · Comparison of Infinitesimals

Basic

Single choice

4 min

Q017 · Recognizing Equivalent Infinitesimals

As $x \rightarrow 0$, which equivalence of infinitesimals is correct?

A. $\ln(1+x) \sim x$

B. $1 - \cos x \sim x$

C. $\sin x \sim x^2$

D. $e^x - 1 \sim 1$

WORKSPACE · add pages as needed

§8 · Continuity and Discontinuities

Basic

Single choice

4 min

Q018 · The Essential Condition for Continuity

Suppose f is defined in a neighborhood of a , including at a . Which statement is equivalent to saying that f is continuous at a ?

- A. $\lim_{x \rightarrow a} f(x) = f(a)$
- B. $f(a)$ exists
- C. The left-hand limit $\lim_{x \rightarrow a^-} f(x)$ exists
- D. f is bounded in some neighborhood of a

WORKSPACE · add pages as needed

§8 · Continuity and
Discontinuities

Basic

Fill in the blank

5 min

Textbook-method adaptation

Q019 · Defining a Missing Value to Obtain Continuity

Let $f(x) = (x^2 - 4)/(x - 2)$ for $x \neq 2$, and define $f(2) = c$. To make f continuous at $x = 2$, one must take $c =$.

WORKSPACE · add pages as needed

§9 · Operations on Continuous Functions

Basic

Single choice

6 min

Q020 · Finding an Interval of Continuity from the Domain

Which set is both the domain and the interval of continuity of $F(x)=\ln(4-x^2)/\sqrt{x+1}$?

- A. A. $(-2,2)$
- B. B. $(-1,2)$
- C. C. $[-1,2)$
- D. D. $(-1,+\infty)$

WORKSPACE · add pages as needed

§9 · Operations on Continuous Functions

Basic

Multiple choice

7 min

Q021 · Algebra Rules for Continuous Functions

Which statements about continuity are correct?

- A. A. Every polynomial is continuous on $\mathbb{R}$
- B. B. A rational function is continuous at every point where its denominator is nonzero
- C. C. A composition of continuous functions is continuous wherever the composition is meaningful
- D. D. The quotient of two continuous functions must remain continuous at a point where the denominator function is zero

WORKSPACE · add pages as needed

§10 · Continuous Functions on Closed Intervals

Basic

True/false with justification

5 min

Q022 · Are Extrema Guaranteed on an Open Interval?

True or false: If f is continuous on an open interval (a,b) , then it must attain both a maximum and a minimum in (a,b) .

WORKSPACE · add pages as needed

Methods

Attempt independently before consulting the solutions.

§1 · Mappings and Functions	Standard	Calculation	8 min	Textbook-method adaptation
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Q023 · A Composite Function and Its Domain

Let $f(x)=\sqrt{x+2}$ and $g(u)=1/(u-1)$. Find the formula and natural domain of $(g\circ f)(x)$.

WORKSPACE · add pages as needed

§1 · Mappings and Functions

Standard

Multiple choice

7 min

Q024 · Analyze Global Properties of a Function

Let $h(x) = |x|/(1+x^2)$ on $\mathbb{R}$. Which statements are correct?

- A. A. h is even
- B. B. For every $x \in \mathbb{R}$, $0 \leq h(x) \leq 1/2$
- C. C. h is periodic
- D. D. h is strictly increasing on $[0, +\infty)$

WORKSPACE · add pages as needed

§1 · Mappings and Functions	Standard	Calculation	8 min	Textbook-method adaptation
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Q025 · An Inverse Function with an Interval Restriction

The domain of $f(x)=2x-3$ is $[1,4]$. Find the range of f , the formula for its inverse, and the domain and range of the inverse.

WORKSPACE · add pages as needed

§1 · Mappings and Functions

Standard

Single choice

6 min

Q026 · Find the Least Positive Period of a Sum

What is the least positive period of $p(x)=\sin(2x)+\cos(4x)$?

- A. $\pi/4$
- B. $\pi/2$
- C. π
- D. 2π

WORKSPACE · add pages as needed

§2 · Limits of Sequences

Standard

Calculation

7 min

Textbook-method adaptation

Q027 · Limit of a Rational Sequence with Equal Degrees

Evaluate $\lim_{n \rightarrow \infty} (5n^2 - 3n + 1) / (2n^2 + n - 4)$.

WORKSPACE · add pages as needed

§2 · Limits of Sequences

Standard

Proof

11 min

Textbook-method adaptation

Q028 · Prove Convergence of a Rational Sequence from the Definition

Use the epsilon-N definition to prove $\lim_{n \rightarrow \infty} (3n-2)/(n+1)=3$.

WORKSPACE · add pages as needed

§3 · Limits of Functions	Standard	Fill in the blank	7 min	Textbook-method adaptation
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Q029 · Determine a Parameter from One-Sided Limits

Let $f(x)=\begin{cases}2x+a & \text{for } x<1; \\ x^2+1 & \text{for } x>1.\end{cases}$ For $\lim_{x\rightarrow 1}f(x)$ to exist, a must equal _____, and the limit is then _____.

WORKSPACE · add pages as needed

§3 · Limits of Functions

Standard

Calculation

7 min

Textbook-method adaptation

Q030 · Use One-Sided Limits to Show Nonexistence

Evaluate $\lim_{x \rightarrow 0} |x|/x$. If it does not exist, explain why.

WORKSPACE · add pages as needed

§3 · Limits of Functions

Standard

Calculation

9 min

Textbook-method adaptation

Q031 · Handle the Sign of a Radical as x Approaches Negative Infinity

Evaluate $\lim_{x \rightarrow -\infty} (2x-1)/\sqrt{x^2+1}$.

WORKSPACE · add pages as needed

§3 · Limits of Functions	Standard	Calculation	6 min	Textbook-method adaptation
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Q032 · Cancel the Common Factor Causing Zero Division

Evaluate $\lim_{x \rightarrow 3} (x^2 - 9)/(x - 3)$.

WORKSPACE · add pages as needed

§4 · Infinitesimals and Infinite
Quantities

Standard

Fill in the blank

6 min

Textbook-method adaptation

Q033 · Writing a Function as a Constant Plus an Infinitesimal

As $x \rightarrow 0$, write $f(x) = (3+2x)/(1+x)$ in the form $f(x) = A + \alpha(x)$, where $\alpha(x)$ is infinitesimal. Find A and $\alpha(x)$.

WORKSPACE · add pages as needed

§4 · Infinitesimals and Infinite
Quantities

Standard

Calculation

7 min

Textbook-method adaptation

Q034 · Infinite Behavior of a Reciprocal Square

Find $\lim_{x \rightarrow 0} 1/x^2$, and explain why the reciprocal relationship between $1/x^2$ and x^2 is valid as $x \rightarrow 0$.

WORKSPACE · add pages as needed

§5 · Algebra of Limits

Standard

Calculation

7 min

Textbook-method adaptation

Q035 · Rationalizing with a Conjugate

Evaluate $\lim_{x \rightarrow 1} (\sqrt{x+3}-2)/(x-1)$ without derivatives.

WORKSPACE · add pages as needed

§5 · Algebra of Limits

Standard

Calculation

8 min

Textbook-method adaptation

Q036 · Algebraic Simplification of a Cube-Root Difference

Evaluate $\lim_{x \rightarrow 8} (\sqrt[3]{x} - 2)/(x - 8)$ without derivatives.

WORKSPACE · add pages as needed

§5 · Algebra of Limits

Standard

Fill in the blank

6 min

Q037 · A Rational Expression from Known Limits

If $f(x) \rightarrow 2$ and $g(x) \rightarrow -1$ as $x \rightarrow a$, then $\lim_{x \rightarrow a} [3f(x) - g(x)] / [f(x) - g(x)] = \underline{\hspace{2cm}}$.

WORKSPACE · add pages as needed

§5 · Algebra of Limits

Standard

True/false with justification

7 min

Q038 · A Quotient of Two Infinitesimals

True or false, with justification: If $\alpha(x) \rightarrow 0$ and $\beta(x) \rightarrow 0$, then $\alpha(x)/\beta(x) \rightarrow 1$.

WORKSPACE · add pages as needed

§5 · Algebra of Limits

Standard

Multiple choice

7 min

Textbook-method adaptation

Q039 · A Rational Limit at Negative Infinity

For $L = \lim_{x \rightarrow -\infty} (3x^2 - x + 1)/(2x^2 + 5)$, which statements are correct? Select all that apply.

- A. A. $L = 3/2$
- B. B. Replacing $x \rightarrow -\infty$ by $x \rightarrow +\infty$ still gives $3/2$
- C. C. After division by x^2 , both $1/x$ and $1/x^2$ tend to 0
- D. D. Since $x \rightarrow -\infty$, $x^2 \rightarrow -\infty$

WORKSPACE · add pages as needed

§6 · Existence Criteria and Two Fundamental Limits	Standard	Calculation	7 min	Textbook-method adaptation
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Q040 · Standardizing Two Sine Factors

Evaluate $\lim_{x \rightarrow 0} [\sin(2x)\sin(3x)]/x^2$.

WORKSPACE · add pages as needed

§6 · Existence Criteria and
Two Fundamental Limits

Standard

Calculation

8 min

Textbook-method adaptation

Q041 · Deriving a Cosine Limit from the First Fundamental Limit

Evaluate $\lim_{x \rightarrow 0} [1 - \cos(2x)]/x^2$ without Taylor expansions.

WORKSPACE · add pages as needed

§6 · Existence Criteria and Two Fundamental Limits	Standard	Calculation	8 min	Textbook-method adaptation
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Q042 · Creating the Second Fundamental Limit

Evaluate $\lim_{x \rightarrow 0} (1+2x)^{3/x}$.

WORKSPACE · add pages as needed

§6 · Existence Criteria and
Two Fundamental Limits

Standard

Calculation

8 min

Textbook-method adaptation

Q043 · An Exponential Sequence with a Negative Increment

Evaluate $\lim_{n \rightarrow \infty} (1 - 3/n)^n$.

WORKSPACE · add pages as needed

§6 · Existence Criteria and Two Fundamental Limits	Standard	Multiple choice	7 min
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Q044 · Recognizing Common Indeterminate Forms

Which limit structures are indeterminate forms? Select all that apply.

- A. $0/0$
- B. ∞/∞
- C. 1^∞
- D. 0^1

WORKSPACE · add pages as needed

§7 · Comparison of Infinitesimals	Standard	Fill in the blank	5 min
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Q045 · Comparing Orders of Power Infinitesimals

As $x \rightarrow 0$, $|x|^3$ is a ____-order infinitesimal relative to x^2 because $|x|^3/x^2 \rightarrow$ ____.

WORKSPACE · add pages as needed

§7 · Comparison of
Infinitesimals

Standard

Calculation

8 min

Textbook-method adaptation

Q046 · Equivalent Substitution in a Product-Quotient Structure

Evaluate $\lim_{x \rightarrow 0} [\sin(3x) \cdot \ln(1+2x)] / [x(e^x - 1)]$.

WORKSPACE · add pages as needed

§7 · Comparison of Infinitesimals	Standard	Calculation	8 min	Textbook-method adaptation
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Q047 · Balancing Second- and First-Order Factors

Evaluate $\lim_{x \rightarrow 0} [1 - \cos(2x)] / [x \sin(3x)]$.

WORKSPACE · add pages as needed

§8 · Continuity and Discontinuities	Standard	True/false with justification	5 min
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Q048 · Can a Removable Discontinuity Always Be Repaired?

True or false: If $x=a$ is a removable discontinuity of f , then redefining only $f(a)$ can always make f continuous at a .

WORKSPACE · add pages as needed

§8 · Continuity and Discontinuities	Standard	Calculation	12 min	Textbook-method adaptation
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Q049 · Parameters for Continuity at Two Junctions

Let $f(x)=ax+b$ for $x<1$, $f(x)=x^2+1$ for $1\leq x\leq 2$, and $f(x)=3x-a$ for $x>2$. Find a,b so that f is continuous on $\mathbb{R}$.

WORKSPACE · add pages as needed

§8 · Continuity and Discontinuities	Standard	Multiple choice	8 min
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Q050 · Distinguishing Types of Discontinuity

Which statements about a discontinuity at $x=a$ are correct?

- A. A. If both one-sided limits exist, are equal and finite, but their common value differs from $f(a)$ (or $f(a)$ is undefined), then a is a removable discontinuity
- B. B. If both one-sided limits are finite but unequal, then a is a jump discontinuity
- C. C. If at least one one-sided limit is infinite, then a is an infinite discontinuity and is of the second kind
- D. D. If the limit fails to exist because of oscillation, then a is a discontinuity of the first kind

WORKSPACE · add pages as needed

§8 · Continuity and Discontinuities	Standard	Proof	14 min	Textbook-method adaptation
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Q051 · Proving Continuity of a Quadratic from the Definition

Without citing the theorem that polynomials are continuous, use the ϵ - δ definition to prove that $f(x)=x^2+1$ is continuous at every $a\in\mathbb{R}$.

WORKSPACE · add pages as needed

§9 · Operations on Continuous Functions	Standard	Calculation	9 min	Textbook-method adaptation
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Q052 · Evaluating a Limit by Continuity of Elementary Functions

Use continuity to evaluate $\lim_{x \rightarrow \pi/6} (\ln(2+\cos x))/(\sqrt{1+\sin x})$, and explain why direct substitution is valid.

WORKSPACE · add pages as needed

§9 · Operations on Continuous Functions	Standard	Proof	15 min	Textbook-method adaptation
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Q053 · The Square Root of a Positive Continuous Function

Suppose f is continuous at a and $f(a)>0$. Prove that in some neighborhood of a , $f(x)>f(a)/2>0$, and then prove that $g(x)=\sqrt{f(x)}$ is continuous at a .

WORKSPACE · add pages as needed

§10 · Continuous Functions on Closed Intervals	Standard	Fill in the blank	6 min	Textbook-method adaptation
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Q054 · A Value Between the Endpoint Values

Suppose f is continuous on $[0,2]$, with $f(0)=-3$ and $f(2)=5$. By the intermediate value theorem, there is at least one $\xi \in$ such that $f(\xi)=1$.

WORKSPACE · add pages as needed

§10 · Continuous Functions on Closed Intervals	Standard	Calculation	14 min	Textbook-method adaptation
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Q055 · Locating and Proving Uniqueness of a Cubic Root

Without using derivatives, prove that $x^3+x-1=0$ has exactly one real root in $(1/2,3/4)$.

WORKSPACE · add pages as needed

Synthesis

Attempt independently before consulting the solutions.

§1 · Mappings and Functions	Advanced	Proof	10 min
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Q056 · Parity of a Composite Function

Let D be symmetric about the origin. Suppose $g:D \rightarrow E$ is even, $f:E \rightarrow R$ is arbitrary, and $h=f \circ g$ is defined on D . Prove that h is even on D .

WORKSPACE · add pages as needed

§1 · Mappings and Functions	Advanced	Parameter / synthesis / application	14 min	Textbook-method adaptation
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Q057 · A Piecewise Parking-Fee Model

A parking lot charges by continuous parking time t in hours. For $0 < t \leq 2$, the fee is 6 yuan. For $2 < t \leq 5$, it charges 6 yuan plus 3 yuan per hour for the portion beyond 2 hours. Write the piecewise formula $C(t)$, find its domain and range, and describe its monotonicity and boundedness.

WORKSPACE · add pages as needed

§2 · Limits of Sequences	Advanced	Calculation	10 min	Textbook-method adaptation
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Q058 · Rationalize a Difference of Radicals

Evaluate $\lim_{n \rightarrow \infty} [\sqrt{n^2+3n}-n]$.

WORKSPACE · add pages as needed

§2 · Limits of Sequences

Advanced

Proof

10 min

Textbook-method adaptation

Q059 · Eventual Positivity from a Positive Limit

Suppose $a_n \rightarrow a$ and $a > 0$. Prove that there exists a positive integer N such that $a_n > a/2$ whenever $n > N$.

WORKSPACE · add pages as needed

§2 · Limits of Sequences	Advanced	Multiple choice	8 min
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Q060 · Necessary Consequences of a Known Limit

Suppose $a_n \rightarrow 2$. Which conclusions must be true?

- A. $\{a_n\}$ is bounded
- B. There exists N such that $a_n > 0$ whenever $n > N$
- C. There exists N such that the tail a_n is strictly increasing
- D. Every subsequence of $\{a_n\}$ converges to 2

WORKSPACE · add pages as needed

§3 · Limits of Functions	Advanced	Proof	14 min	Textbook-method adaptation
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Q061 · An Epsilon-Delta Proof for a Quadratic Limit

Use the epsilon-delta definition to prove $\lim_{x \rightarrow 2} (x^2 + 1) = 5$.

WORKSPACE · add pages as needed

§3 · Limits of Functions

Advanced

Fill in the blank

7 min

Textbook-method adaptation

Q062 · Opposite Signs Near a Vertical Blow-Up

Fill in the blanks: $\lim_{x \rightarrow 2^-} 1/(x-2) = \underline{\hspace{2cm}}$, $\lim_{x \rightarrow 2^+} 1/(x-2) = \underline{\hspace{2cm}}$, and therefore $\lim_{x \rightarrow 2} 1/(x-2) \underline{\hspace{2cm}}$.

WORKSPACE · add pages as needed

§3 · Limits of Functions	Advanced	Calculation	10 min	Textbook-method adaptation
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Q063 · Limit of a Difference of Two Radicals

Evaluate $\lim_{x \rightarrow 0} [\sqrt{1+x} - \sqrt{1-x}]/x$.

WORKSPACE · add pages as needed

§4 · Infinitesimals and Infinite Quantities	Advanced	Calculation	14 min
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Q064 · Parameter Classification with an Oscillatory Factor

Let $f_p(x)=|x|^p \sin(1/x)$ for $x\neq 0$. For $p>0$, $p=0$, and $p<0$, decide whether f_p is infinitesimal, bounded with no limit, or unbounded without tending to $+\infty$ or $-\infty$ as $x\rightarrow 0$.

WORKSPACE · add pages as needed

§4 · Infinitesimals and Infinite Quantities

Advanced

True/false with justification

10 min

Q065 · Does Local Unboundedness Imply an Infinite Limit?

True or false, with justification: If f is unbounded in every deleted neighborhood of $x=0$, then $\lim_{x \rightarrow 0} |f(x)| = +\infty$.

WORKSPACE · add pages as needed

§5 · Algebra of Limits	Advanced	Calculation	14 min
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Q066 · Double Rationalization of Nested Radicals

Evaluate $\lim_{x \rightarrow 0} [\sqrt{1 + \sqrt{1 + x}} - \sqrt{2}] / x$ without derivatives or Taylor expansions.

WORKSPACE · add pages as needed

§5 · Algebra of Limits	Advanced	Calculation	14 min
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Q067 · Determining Parameters from a Finite Limit

Find real numbers a, b such that $\lim_{x \rightarrow 1} (x^2 + ax + b) / (x - 1) = 3$.

WORKSPACE · add pages as needed

§5 · Algebra of Limits	Advanced	Calculation	12 min
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Q068 · Using a Known Ratio of Deviations

As $x \rightarrow a$, suppose $f(x) \rightarrow 1$, $g(x) \rightarrow 2$, and $[f(x)-1]/[g(x)-2] \rightarrow 3$. Find $\lim_{x \rightarrow a} [f(x)^2-1]/[g(x)^2-4]$.

WORKSPACE · add pages as needed

§5 · Algebra of Limits	Advanced	Proof	20 min	Textbook-method adaptation
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Q069 · Proving the Product Limit Theorem from Definitions

Suppose $f(x) \rightarrow A$ and $g(x) \rightarrow B$ as $x \rightarrow a$. Using only the limit definition and local boundedness of convergent functions, prove that $f(x)g(x) \rightarrow AB$.

WORKSPACE · add pages as needed

§6 · Existence Criteria and
Two Fundamental Limits

Advanced

Calculation

8 min

Textbook-method adaptation

Q070 · A Ratio of Sines with Different Scales

Evaluate $\lim_{x \rightarrow 0} \sin(3x)/\sin(5x)$.

WORKSPACE · add pages as needed

§6 · Existence Criteria and Two Fundamental Limits	Advanced	Calculation	12 min	Textbook-method adaptation
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Q071 · The Second Fundamental Limit with a Fractional Base

Evaluate $\lim_{x \rightarrow 0} [(1+x)/(1-x)]^{1/x}$.

WORKSPACE · add pages as needed

§6 · Existence Criteria and Two Fundamental Limits	Advanced	Calculation	12 min
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Q072 · Determining Parameters from an Exponential Limit

Let a, b be positive real numbers. Given $\lim_{x \rightarrow 0} (1+ax)^{b/x} = e^6$ and $a+b=5$, find all ordered pairs (a, b) .

WORKSPACE · add pages as needed

§6 · Existence Criteria and Two Fundamental Limits	Advanced	Proof	14 min	Textbook-method adaptation
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Q073 · Squeezing the Floor-Function Error

Use the squeeze theorem to prove $\lim_{x \rightarrow +\infty} \lfloor x \rfloor / x = 1$.

WORKSPACE · add pages as needed

§6 · Existence Criteria and
Two Fundamental Limits

Advanced

Proof

18 min

Textbook-method adaptation

Q074 · A Monotone-Bounded Proof for a Linear Recurrence

Let $a_1 > 2$ and $a_{n+1} = (a_n + 2)/3$. Prove that $\{a_n\}$ converges and find its limit, without first deriving a closed form.

WORKSPACE · add pages as needed

§7 · Comparison of Infinitesimals	Advanced	Calculation	11 min
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Q075 · A Mixed Exponential-Logarithmic-Trigonometric Limit

Evaluate $\lim_{x \rightarrow 0} [(e^{2x}-1)\tan(3x)]/[x \ln(1+4x)]$.

WORKSPACE · add pages as needed

§7 · Comparison of Infinitesimals	Advanced	Calculation	12 min
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Q076 · Determining Parameters from an Equivalence

Let $a, b > 0$. If $e^{ax} - 1 \sim b \sin(2x)$ as $x \rightarrow 0$ and $a + b = 3$, find a and b .

WORKSPACE · add pages as needed

§7 · Comparison of Infinitesimals	Advanced	Calculation	12 min
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Q077 · Checking a First-Order Remainder in a Difference

Without Taylor expansions, evaluate $\lim_{x \rightarrow 0} [\sqrt{(1+x)} - 1 - x/2]/x$.

WORKSPACE · add pages as needed

§8 · Continuity and Discontinuities	Advanced	Calculation	16 min	Textbook-method adaptation
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Q078 · Three Types of Discontinuity in One Function

Let $f(x)=-1$ for $x<0$; $f(x)=(x^2-1)/(x-1)$ for $0\leq x<2$ with $x\neq 1$; and $f(x)=1/(x-2)$ for $x>2$. Find and classify every discontinuity. If one is removable, give the value for a continuous extension.

WORKSPACE · add pages as needed

§8 · Continuity and Discontinuities	Advanced	Error diagnosis	14 min
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Q079 · Must the Sum of Two Discontinuous Functions Be Discontinuous?

A student claims: “If both f and g are discontinuous at $x=0$, then $f+g$ must be discontinuous at 0 , because only sums of continuous functions are continuous.” Identify the logical error and construct and verify a counterexample.

WORKSPACE · add pages as needed

§8 · Continuity and Discontinuities	Advanced	Proof	20 min	Textbook-method adaptation
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Q080 · Finding the Unique Junction Value by One-Sided Continuity

Fix a real number a . Define $f(x)=\sin(x-a)/(x-a)$ for $x<a$, $f(a)=c$, and $f(x)=\ln(1+x-a)/(x-a)$ for $x>a$. First prove that a piecewise function is continuous at a exactly when both one-sided limits equal $f(a)$, and then find the unique c making this function continuous.

WORKSPACE · add pages as needed

§10 · Continuous Functions on Closed Intervals	Advanced	Calculation	16 min	Textbook-method adaptation
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Q081 · Finding Closed-Interval Extrema without Derivatives

Without using derivatives, find the maximum and minimum of $f(x)=x^4-2x^2+3$ on $[-2,2]$, together with all points where they are attained.

WORKSPACE · add pages as needed

§10 · Continuous Functions on Closed Intervals	Advanced	Calculation	18 min
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Q082 · Finding a Complete Range from Bounds and the Intermediate Value Property

Without using derivatives, find the range of $f(x)=x+4/(x+1)$ on $[0,3]$, and explain why every number between the two bounds is attained.

WORKSPACE · add pages as needed

§1 · Mappings and Functions

Hard

Error diagnosis

12 min

Textbook-method adaptation

Q083 · Diagnose an Endpoint Error in a Domain

Find the domain of $F(x) = \ln(1 - \sqrt{x-1})$. A student writes: “The radical requires $x \geq 1$. The logarithm requires $1 - \sqrt{x-1} \geq 0$, so $x \leq 2$. Hence the domain is $[1, 2]$.” Identify the error and give the correct solution.

WORKSPACE · add pages as needed

§1 · Mappings and Functions

Hard

Proof

16 min

Q084 · Inverse of a Composition of Bijections

Let $f:A \rightarrow B$ and $g:B \rightarrow C$ be bijections. Prove that $g \circ f:A \rightarrow C$ is a bijection and that $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

WORKSPACE · add pages as needed

§2 · Limits of Sequences

Hard

Error diagnosis

20 min

Textbook-method adaptation

Q085 · A Limit Equation Cannot Replace a Convergence Proof

Let $a_1=2$ and $a_{n+1}=\sqrt{3+a_n}$. A student writes: “Assume $a_n \rightarrow L$. Then $L=\sqrt{3+L}$, so $L=(1+\sqrt{13})/2$. Therefore the sequence converges to this value.” Identify the logical error and complete a rigorous proof.

WORKSPACE · add pages as needed

§3 · Limits of Functions	Hard	Proof	18 min	Textbook-method adaptation
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Q086 · Prove Uniqueness of a Function Limit

Let a be an accumulation point of the domain of f . If $\lim_{x \rightarrow a} f(x) = A$ and $\lim_{x \rightarrow a} f(x) = B$, prove that $A = B$.

WORKSPACE · add pages as needed

§4 · Infinitesimals and Infinite Quantities

Hard

Proof

18 min

Textbook-method adaptation

Q087 · A Bounded Function Times an Infinitesimal

Suppose $\alpha(x)$ is infinitesimal as $x \rightarrow a$ and $g(x)$ is bounded in a deleted neighborhood of a . Using definitions only, prove that $\alpha(x)g(x)$ is also infinitesimal, treating separately the case where the bound may be 0.

WORKSPACE · add pages as needed

§5 · Algebra of Limits	Hard	Error diagnosis	16 min	Textbook-method adaptation
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Q088 · Diagnosing Errors in 0/0 and Cancellation

A student evaluates $\lim_{x \rightarrow 1} (x^2 - 1)/(x - 1)$ as follows: first, substitution gives $0/0=0$; second, canceling $x-1$ gives $x+1$, so the limit is also 2; hence the answer is both 0 and 2. Identify the logical error in each step and give the complete correct solution.

WORKSPACE · add pages as needed

§6 · Existence Criteria and Two
Fundamental Limits

Hard

Parameter / synthesis / application

18 min

Q089 · A Variable-Exponent Limit at Infinity

Evaluate $\lim_{x \rightarrow +\infty} (1 + 3/x)^{(2x - \sqrt{x})}$, explicitly identifying the 1^∞ form and the effective standardized exponent.

WORKSPACE · add pages as needed

§6 · Existence Criteria and Two Fundamental Limits	Hard	Proof	25 min	Textbook-method adaptation
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Q090 · A Geometric Proof of the First Fundamental Limit

Assuming radian measure, start from $\sin x < x < \tan x$ for $0 < x < \pi/2$ and prove $\lim_{x \rightarrow 0} \sin x/x = 1$, covering both $x \rightarrow 0+$ and $x \rightarrow 0-$.

WORKSPACE · add pages as needed

§7 · Comparison of
Infinitesimals

Hard

Proof

18 min

Textbook-method adaptation

Q091 · Equivalence Characterized by Relative Error

Let $\alpha(x)$ and $\beta(x)$ both be infinitesimal as $x \rightarrow a$, with $\beta(x)$ nonzero in a deleted neighborhood. Prove that $\alpha(x) \sim \beta(x)$ if and only if $\alpha(x) - \beta(x) = o(\beta(x))$.

WORKSPACE · add pages as needed

§7 · Comparison of Infinitesimals	Hard	Proof	20 min	Textbook-method adaptation
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Q092 · Proving Safe Replacement in a Quotient

Suppose $\alpha \sim \alpha_1$ and $\beta \sim \beta_1$ as $x \rightarrow a$, with all relevant quotients defined in a deleted neighborhood. Prove $\alpha/\beta \sim \alpha_1/\beta_1$, and explain why this supports equivalent substitution in products and quotients.

WORKSPACE · add pages as needed

§7 · Comparison of Infinitesimals

Hard

Proof

20 min

Q093 · A Counterexample to Replacement inside a Sum

Construct and verify a counterexample showing that $\alpha \sim \alpha_1$ and $\beta \sim \beta_1$ need not imply $\alpha + \beta \sim \alpha_1 + \beta_1$. Require both sums to be nonzero in a deleted neighborhood.

WORKSPACE · add pages as needed

§8 · Continuity and Discontinuities

Hard

Parameter / synthesis / application

28 min

Q094 · Can a Power Factor Suppress Oscillation?

For a real parameter p , define $f_p(0)=0$ and $f_p(x)=|x|^p \sin(1/x)$ for $x \neq 0$. Discuss continuity at $x=0$. When it is discontinuous, classify the discontinuity and decide whether the function is bounded in some neighborhood of 0.

WORKSPACE · add pages as needed

§10 · Continuous Functions on
Closed Intervals

Hard

Proof

22 min

Textbook-method adaptation

Q095 · Equal Values Half an Interval Apart

Suppose f is continuous on $[0, 1]$ and $f(0) = f(1)$. Prove that there is at least one $\xi \in [0, 1/2]$ such that $f(\xi) = f(\xi + 1/2)$.

WORKSPACE · add pages as needed

§10 · Continuous Functions on Closed Intervals

Hard

Parameter / synthesis / application

30 min

Q096 · Optional: Comparing Uniform Continuity on an Unbounded Interval

On $[0, +\infty)$, determine whether $f(x) = \sqrt{x}$ and $g(x) = x^2$ are uniformly continuous. Prove both conclusions directly from the definition. For the function that is not uniformly continuous, give a specific ε_0 and a contradiction construction.

WORKSPACE · add pages as needed

Challenge

Attempt independently before consulting the solutions.

§2 · Limits of Sequences

Challenge

Parameter / synthesis / application

28 min

Q097 · Characterize Parameter Convergence Using Even and Odd Subsequences

For real p and q , define $a_{2k}=p+1/k$ and $a_{2k-1}=q-1/k$ for $k=1,2,\dots$. Find a necessary and sufficient condition for $\{a_k\}$ to converge. In the convergent case, state the limit and prove it using the epsilon- N definition.

WORKSPACE · add pages as needed

§6 · Existence Criteria and
Two Fundamental Limits

Challenge

Parameter / synthesis /
application

32 min

Textbook-method adaptation

Q098 · A Complete Convergence Chain for a Radical Recurrence

Let $0 < a_1 < 2$ and $a_{n+1} = \sqrt{2 + a_n}$. Prove: (1) $0 < a_n < 2$; (2) $\{a_n\}$ is increasing; (3) it converges; and (4) find its limit. Do not guess a closed form or merely solve a limit equation.

WORKSPACE · add pages as needed

§7 · Comparison of
Infinitesimals

Challenge

Parameter / synthesis /
application

35 min

Textbook-method adaptation

Q099 · Safe and Unsafe Zones for Equivalent Substitution

Complete the following without derivatives, L'Hôpital's rule, or Taylor expansions. (1) Evaluate $A = \lim_{x \rightarrow 0} \sin(2x)/(e^x - 1)$. (2) A student uses $\sqrt{1+x} - 1 \sim x/2$ to replace $\sqrt{1+x} - 1$ inside a difference and claims $B = \lim_{x \rightarrow 0} [\sqrt{1+x} - 1 - x/2]/x^2 = 0$. Judge the method and conclusion, then find B by exact algebra. (3) State one rule describing when direct equivalent-infinitesimal substitution is valid.

WORKSPACE · add pages as needed

§10 · Continuous Functions on Closed Intervals

Challenge

Parameter / synthesis / application

40 min

Q100 · Optional: The Uniform-Continuity Boundary for Oscillatory Functions

On $(0,1]$, define $F(x)=x\sin(1/x)$ and $G(x)=\sin(1/x)$. Determine whether each is uniformly continuous. For F , use a continuous extension and the uniform-continuity theorem on a closed interval. For G , construct two sequences whose distance tends to 0 while their function values differ by exactly 2.

WORKSPACE · add pages as needed

Training value, limitations, and second-pass route

Strengths

- All ten sections are covered and indexed by knowledge point.
- Conceptual, computational, proof, parameter, counterexample, and error-diagnosis tasks expose more than routine algebra.
- Even the hard questions stay within Chapter 1 tools and train the real boundaries of squeeze arguments, equivalent infinitesimals, and continuity theorems.
- Chinese and English identifiers and mathematics match one-to-one, supporting calculus vocabulary development.

Limitations

- One hundred questions cannot exhaust every algebraic transformation; weak categories may need extra drills.
- Textbook material is represented by independently rewritten method adaptations, not verbatim textbook questions.
- Difficulty is partly subjective and depends on prior algebra preparation.
- Uniform continuity is starred enrichment; skip those optional items if your course omits it.
- To respect the Chapter 1 boundary, later shortcuts such as L'Hopital's rule and Taylor expansions are not shown.

Recommended second pass

1. First pass: work under time limits, record confidence, and do not open the solutions.
2. Correction: classify each error as conceptual, algebraic, method-selection, or rigor/communication.
3. After 48 hours: redo wrong items and their neighbors without prompts.
4. After one week: sample by knowledge tag, emphasizing limit definitions, cancellation in equivalent infinitesimals, and theorem hypotheses.